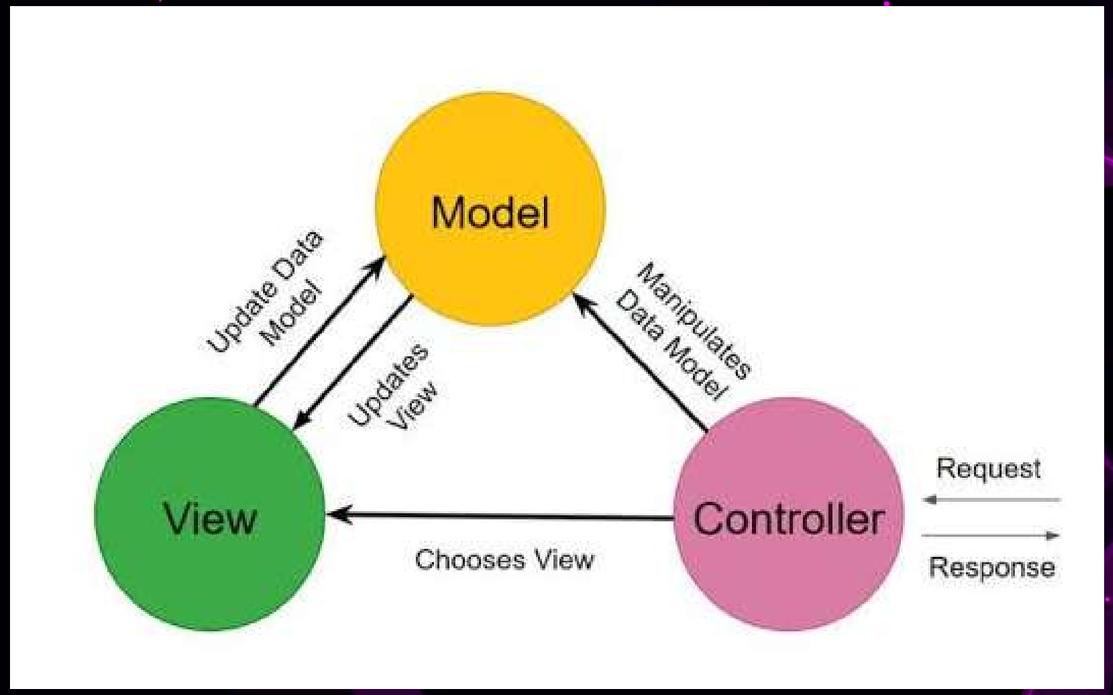
QMS-FLOW SPECIFIC REQUIREMENTS

Presented by QualiTech Solutions

Video link: https://youtu.be/0rick_2ilro?si=GOQK7zWCQQjldzsT

Architectural Rationale

ARCHITECTURE STYLE:
MVC



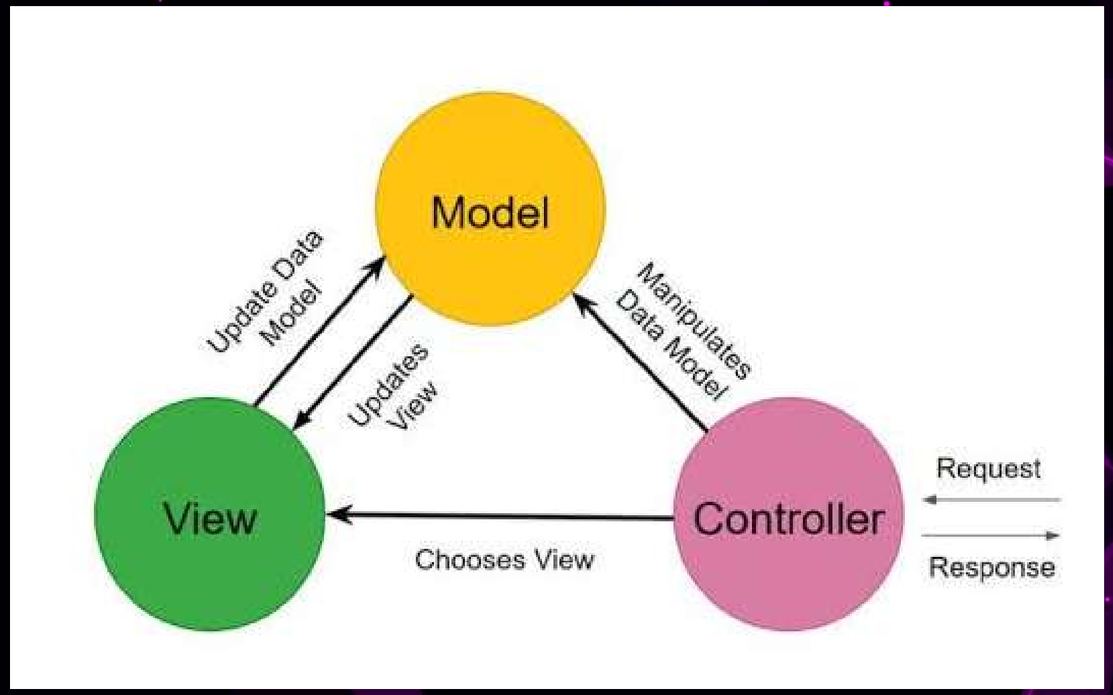
ARCHITECTURE STYLE: MVC

MODEL
- DATA & BUSINESS LOGIC
- AUDIT RECORDS, DOCUMENTS, CAPA

VIEW
- USER INTERFACE
- ROLE-BASED DASHBOARDS

CONTROLLER
- PROCESSES USER INPUT
- COORDINATES MODEL & VIEW

Architectural Rationale



WHY WE CHOSE MVC

1. AGILE DEVELOPMENT SUPPORT
   - PARALLEL FRONTEND/BACKEND WORK
   - ITERATIVE DELIVERY

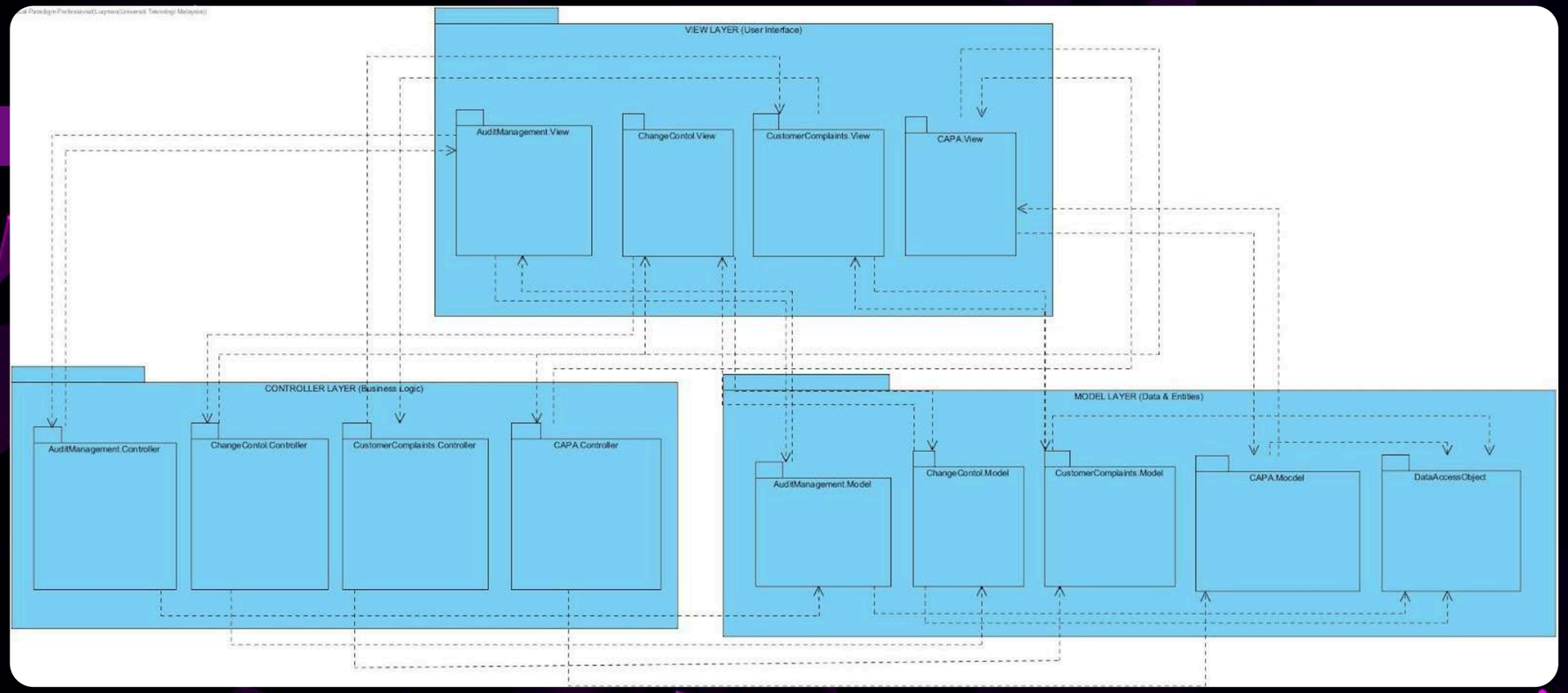
2. ROLE-BASED CUSTOMIZATION
   - TAILORED INTERFACES PER USER ROLE
   - UNIFIED DATA MODEL

3. MAINTAINABILITY
   - INDEPENDENT LAYER UPDATES
   - EASY TESTING

4. TECHNOLOGY ALIGNMENT
   - COMPATIBLE WITH SPRING BOOT & REACT.JS
   - MODERN WEB STANDARDS

5. ENTERPRISE READINESS
   - ACTIVE DIRECTORY INTEGRATION
   - MICROSOFT 365 COMPATIBILITY

Implemented View: Architectural Diagram



1. View Layer (User Interface):

This topmost layer contains web-based user interfaces for all four core modules: Audit Management, Change Control, Customer Complaints, and Corrective Actions (CAPA). Each module provides role-specific UI components (e.g., AuditUI, DocumentUI, ComplaintUI) accessible via modern web browsers. This layer communicates with external systems, including Active Directory for authentication and email services for notifications.

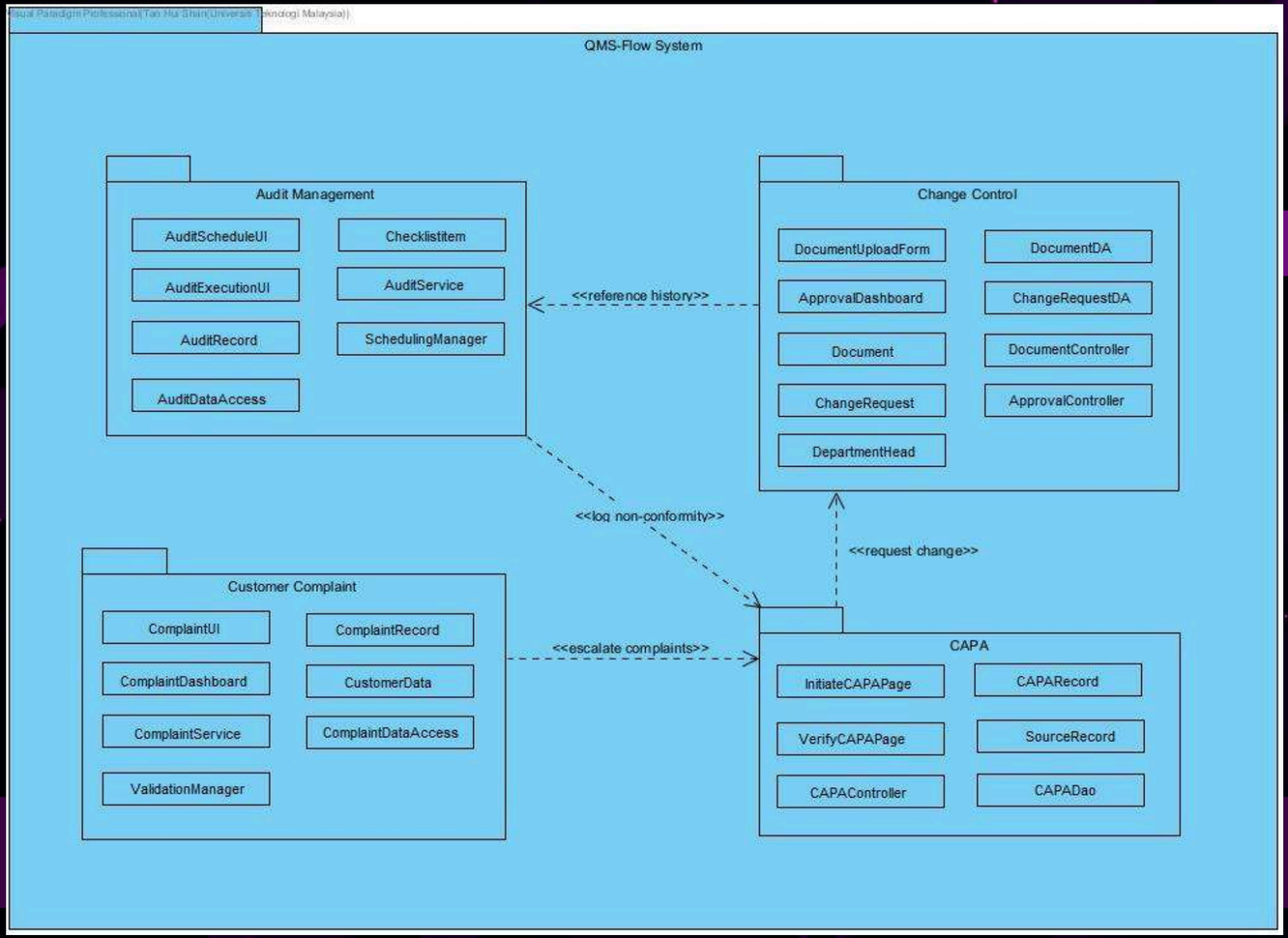
2. Controller Layer (Business Logic):

The middle layer implements the business logic and workflow management for each module through dedicated controller components (e.g., AuditController, DocumentController). These controllers process user requests, enforce business rules, manage ISO-compliant workflows, and coordinate data flow between the View and Model layers.

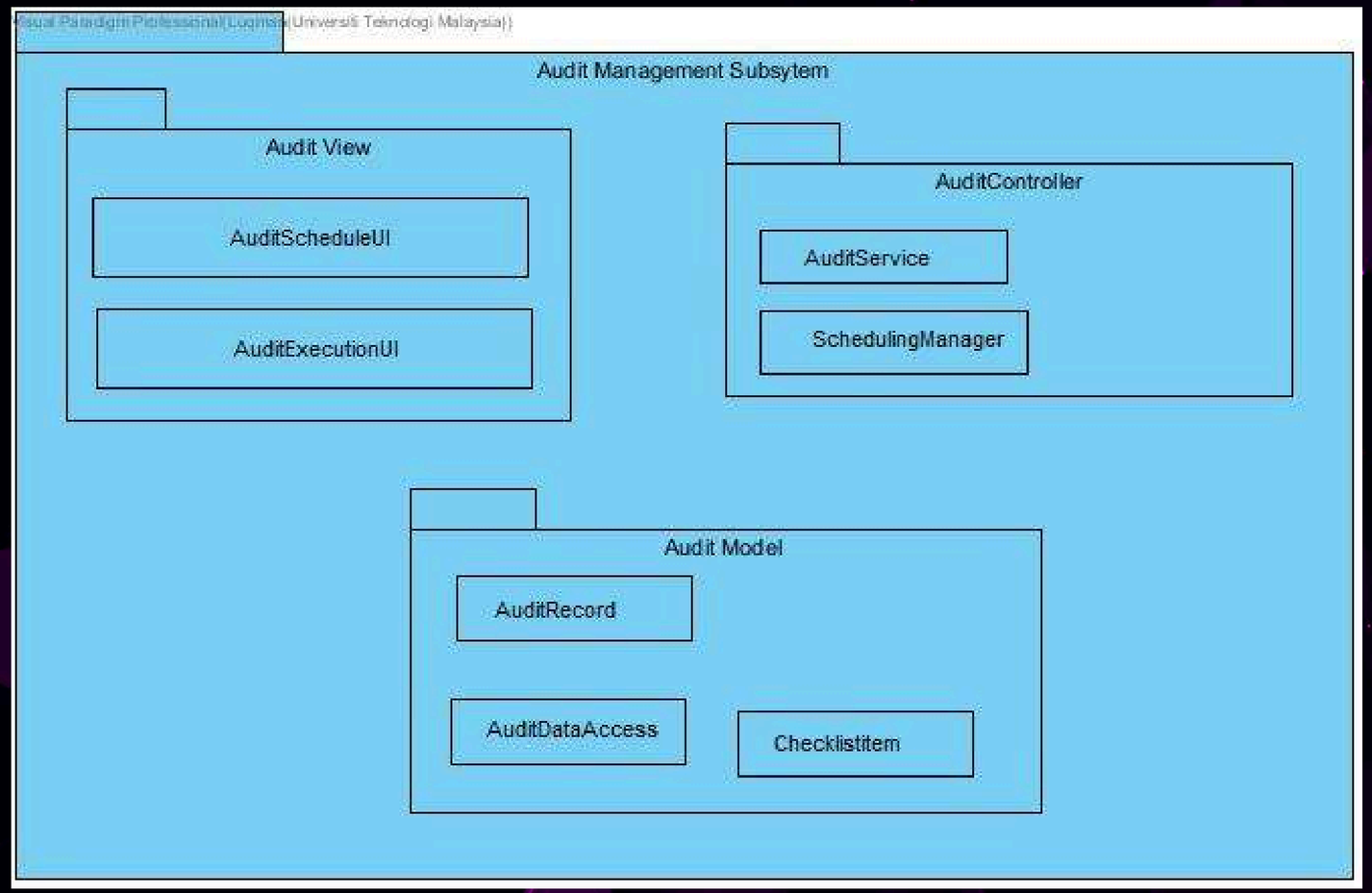
3. Model Layer (Data & Entities):

The foundation layer represents the core business entities (e.g., Audit, Document, Complaint, CAPARecord) and handles all data persistence operations. It interfaces with a PostgreSQL database for structured data storage and maintains complete audit trails as required by ISO 9001:2015 standards.

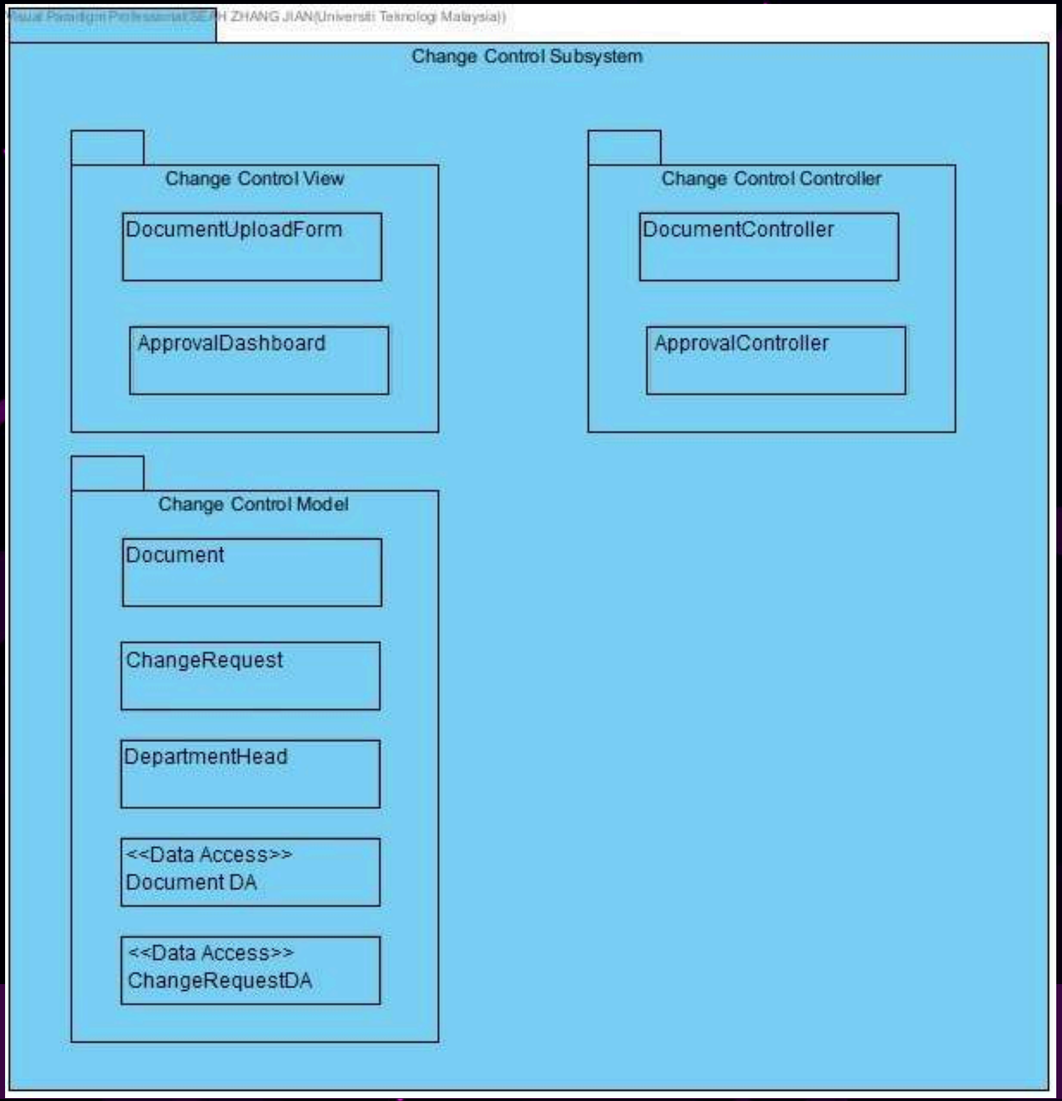
Implementation View: Complete Package Diagram



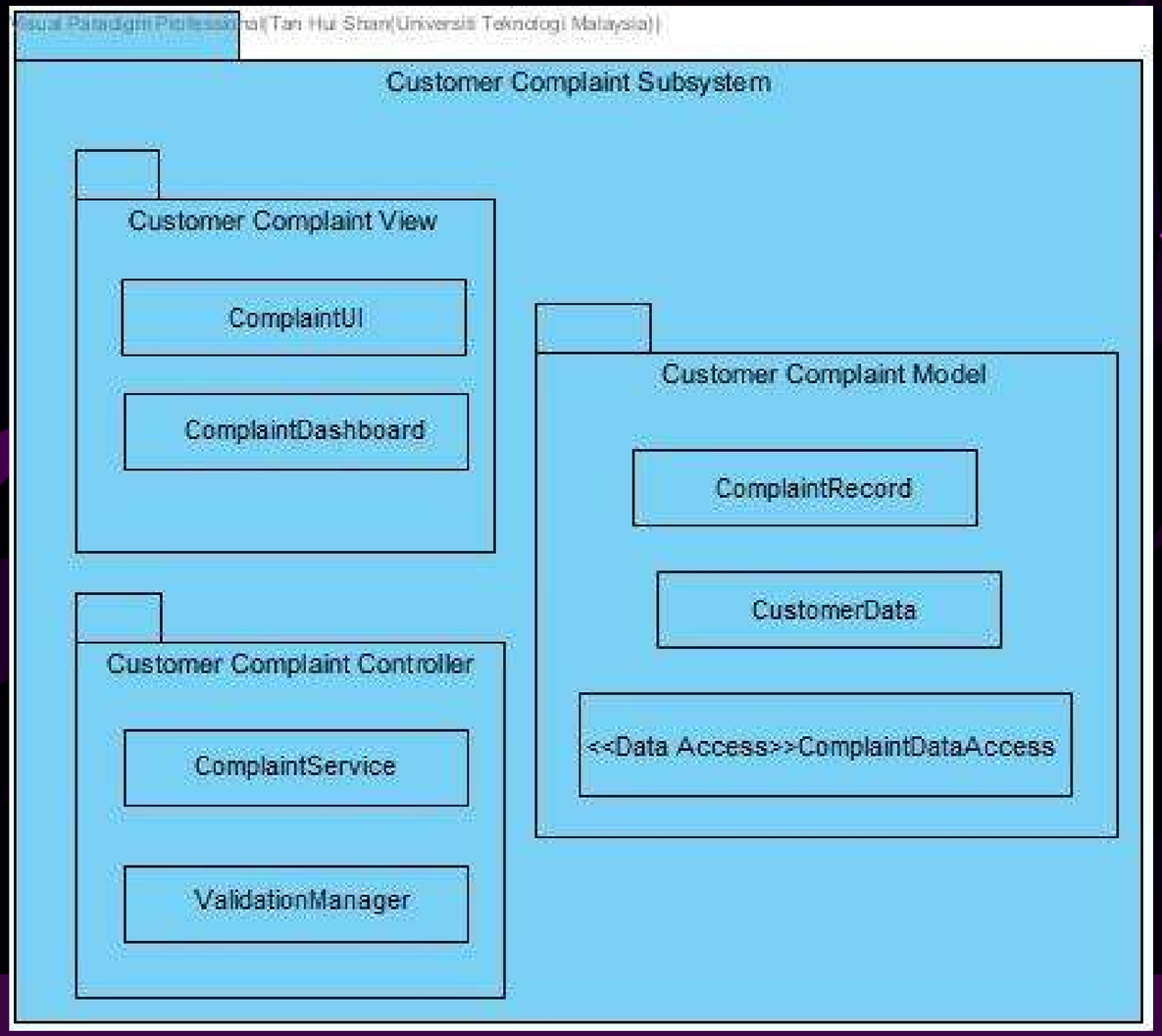
Audit Management Subsystem



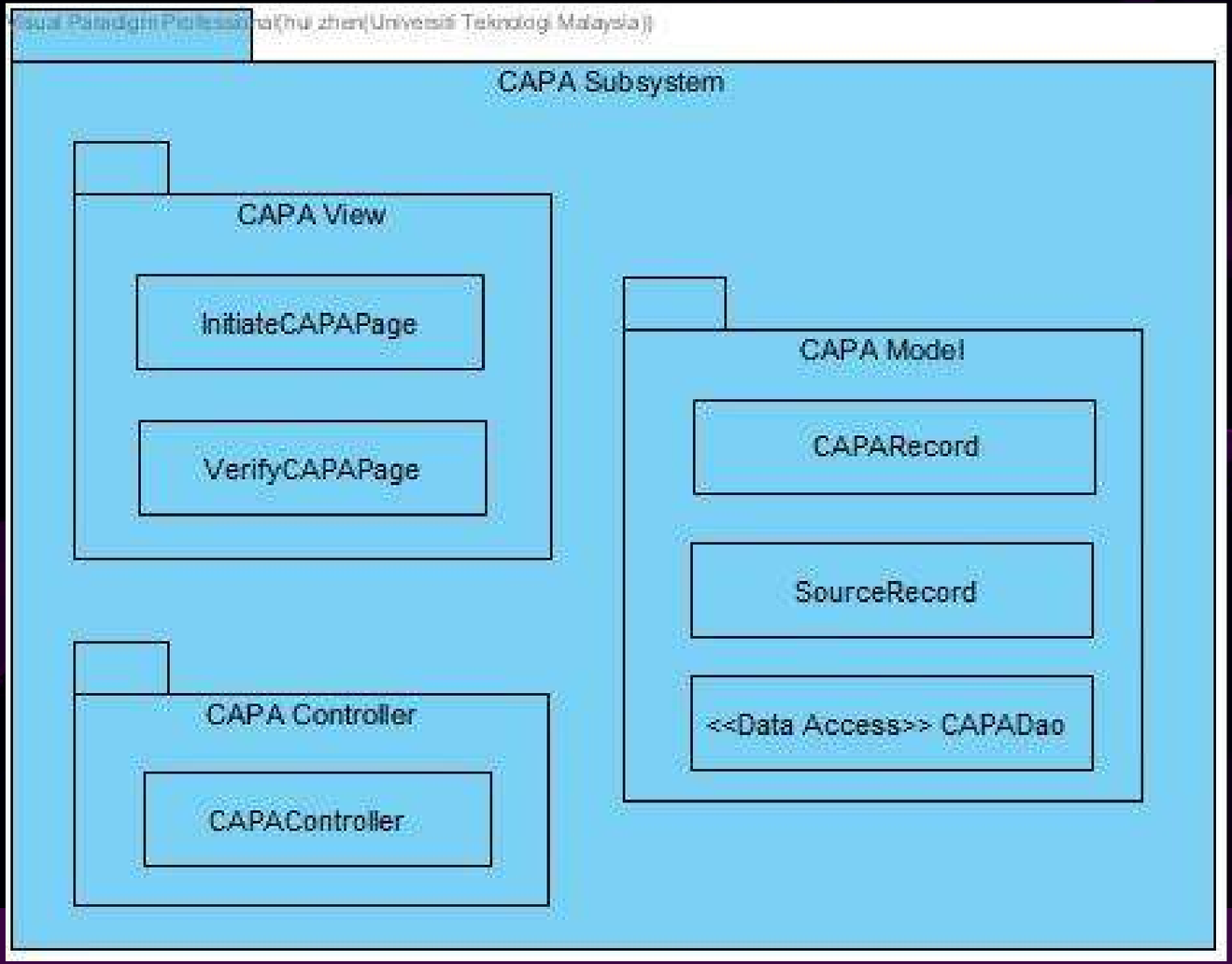
Change Control Subsystem



Customer Complaint Subsystem



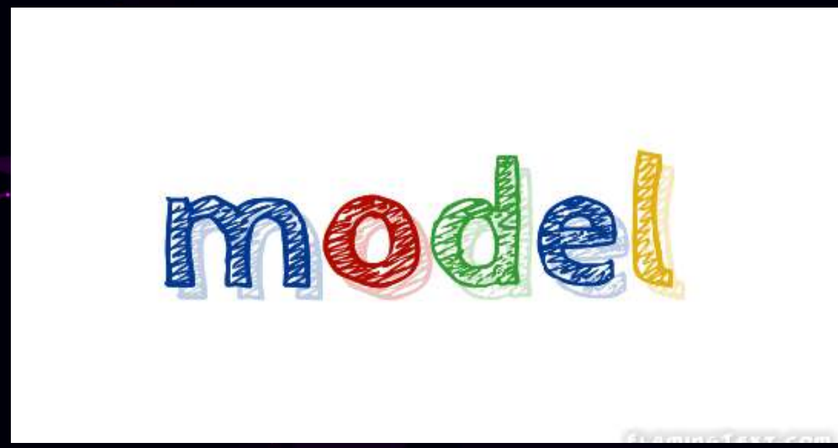
CAPA Subsystem



Logical View: Class Diagram

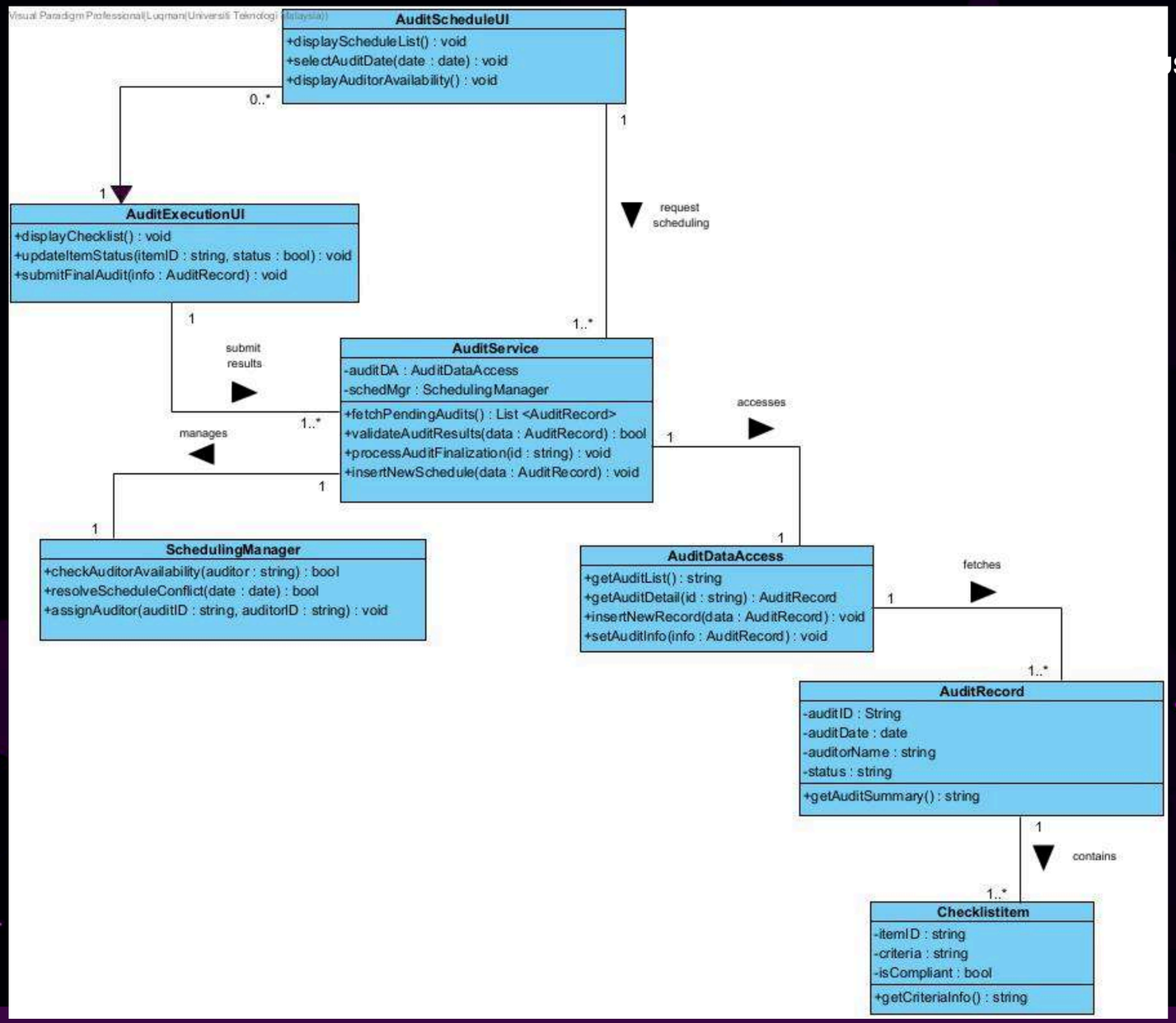
View

Controller

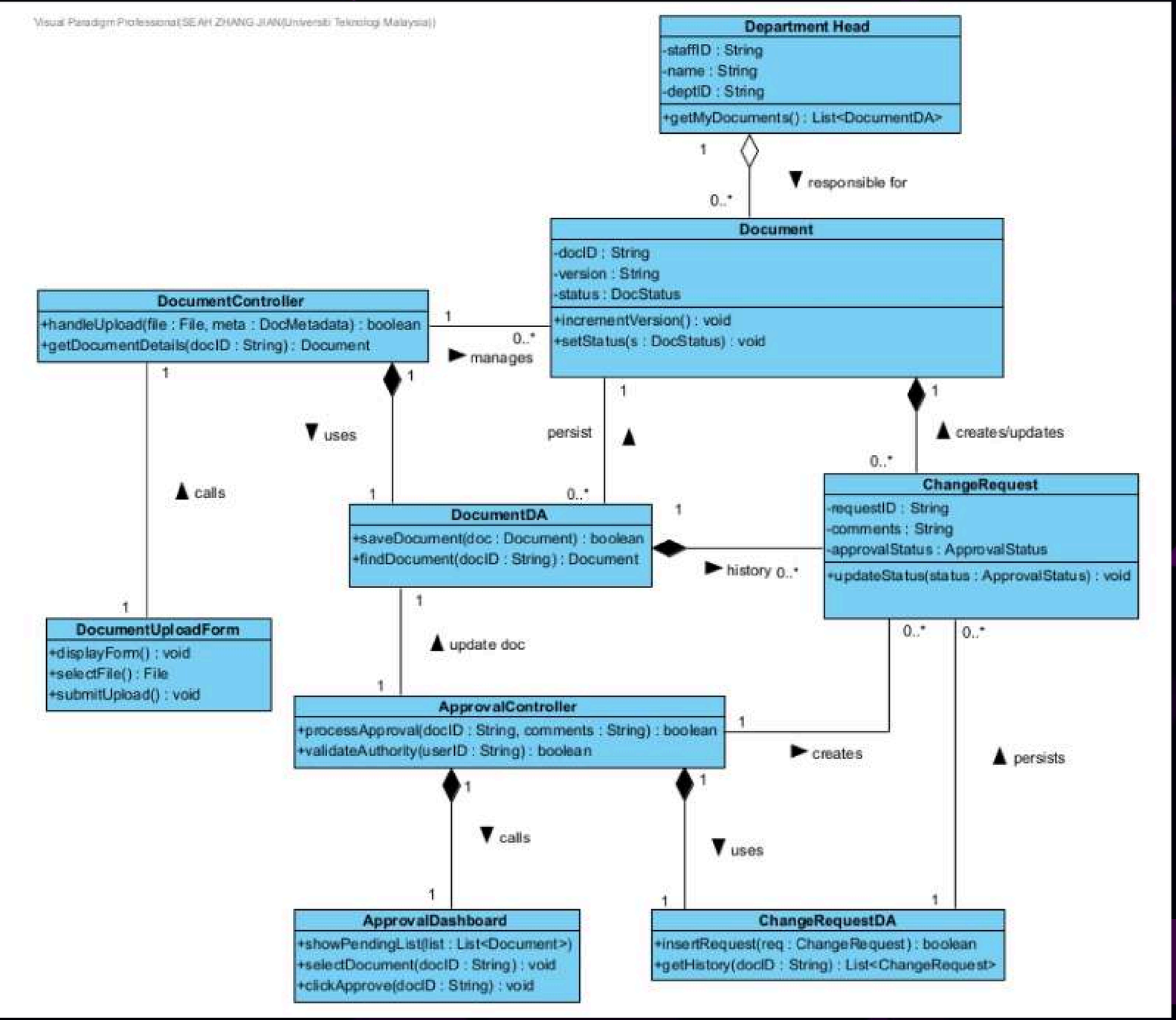


Model

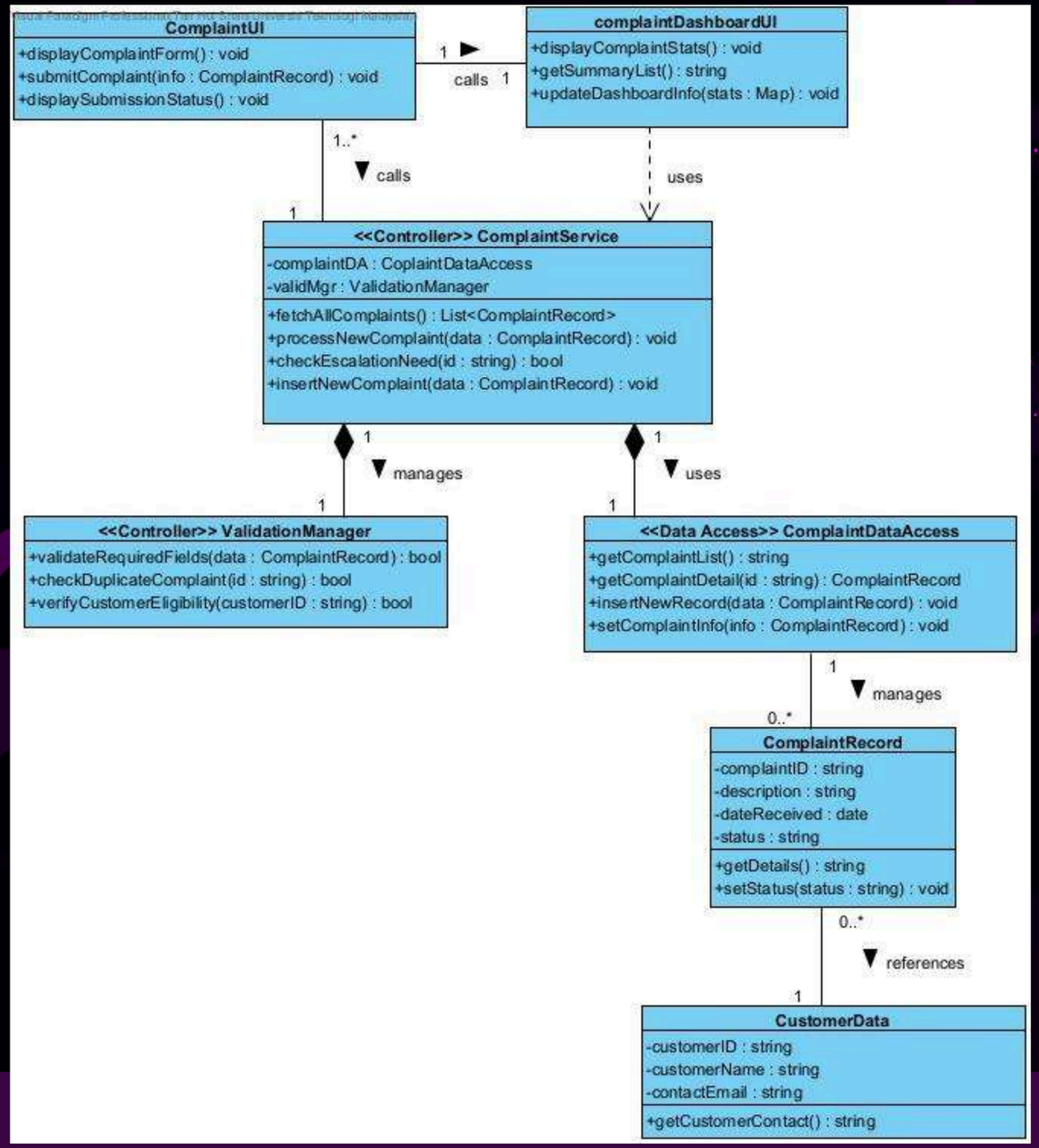
Class Diagram: <Audit Management> Subsystem



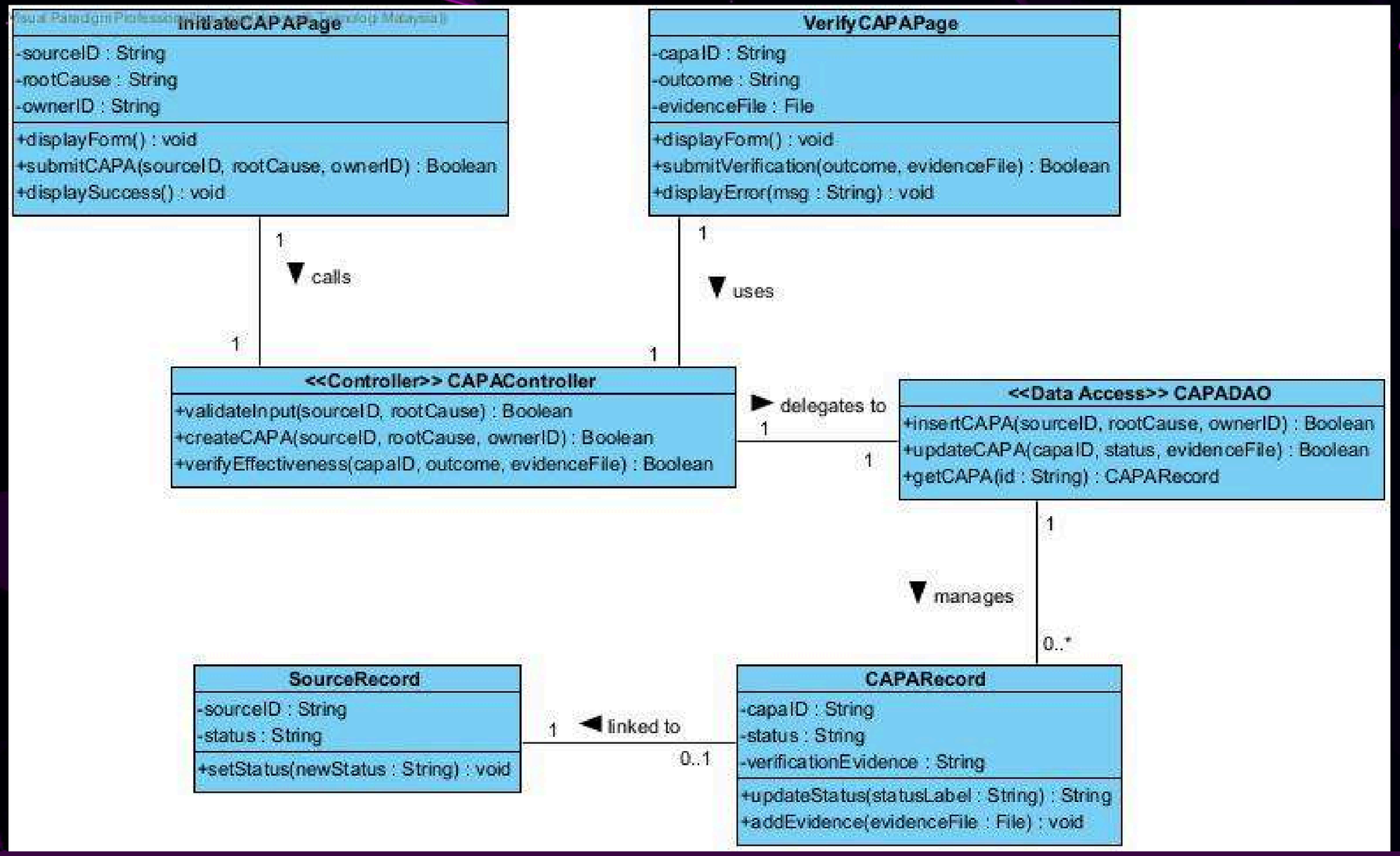
Class Diagram: <Change Control> Subsystem



Class Diagram: <Customer Complaint> Subsystem



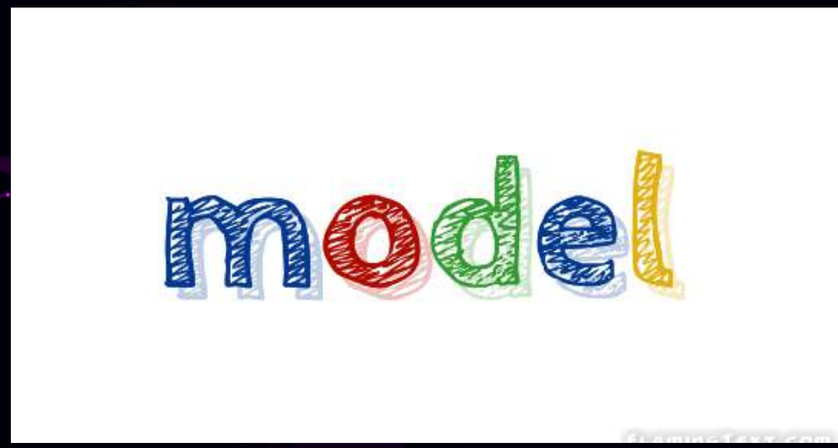
Class Diagram: <CAPA Management> Subsystem



Process View: Sequence Diagram

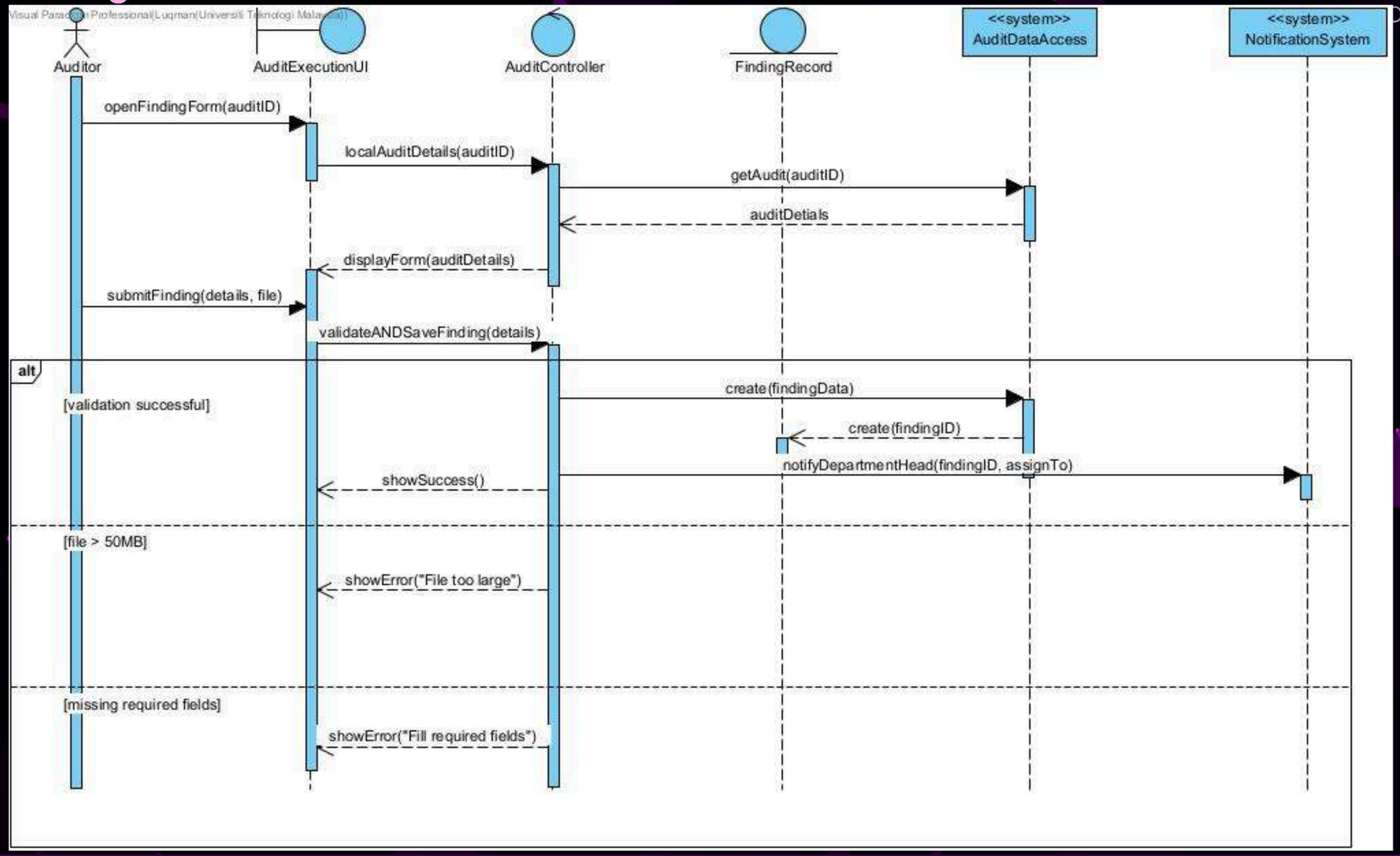
View

Controller

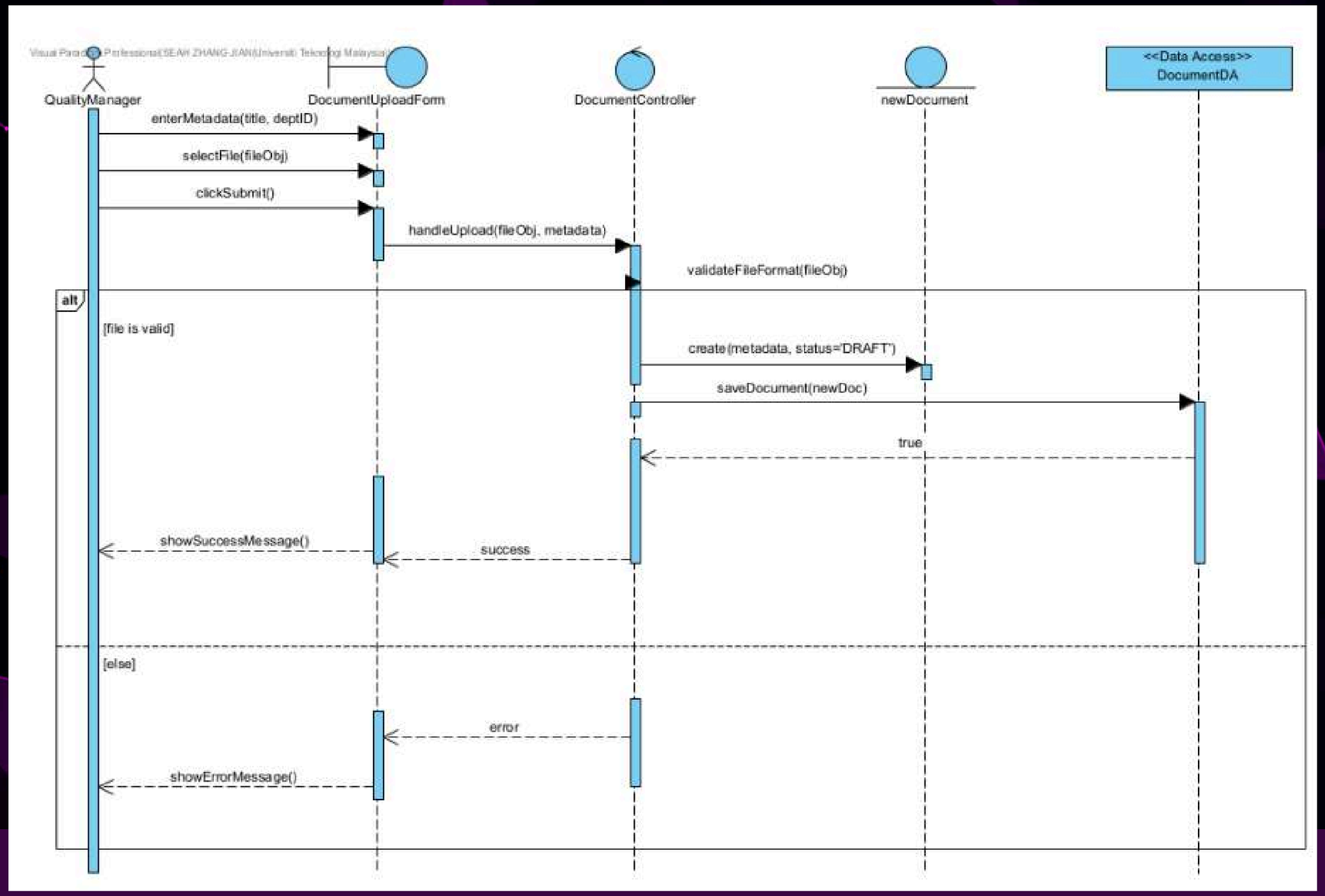


Model

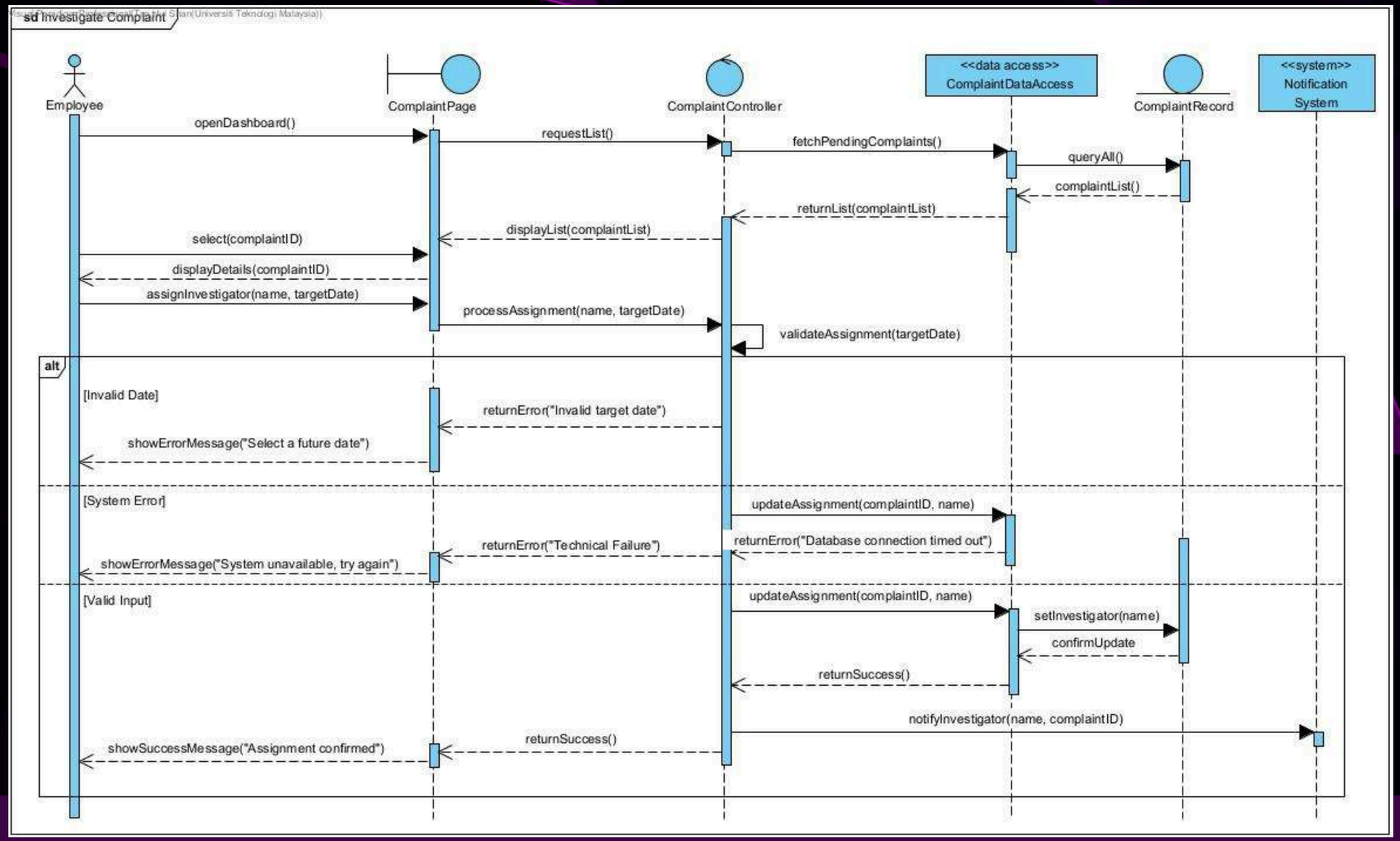
Sequence Diagram: SD002



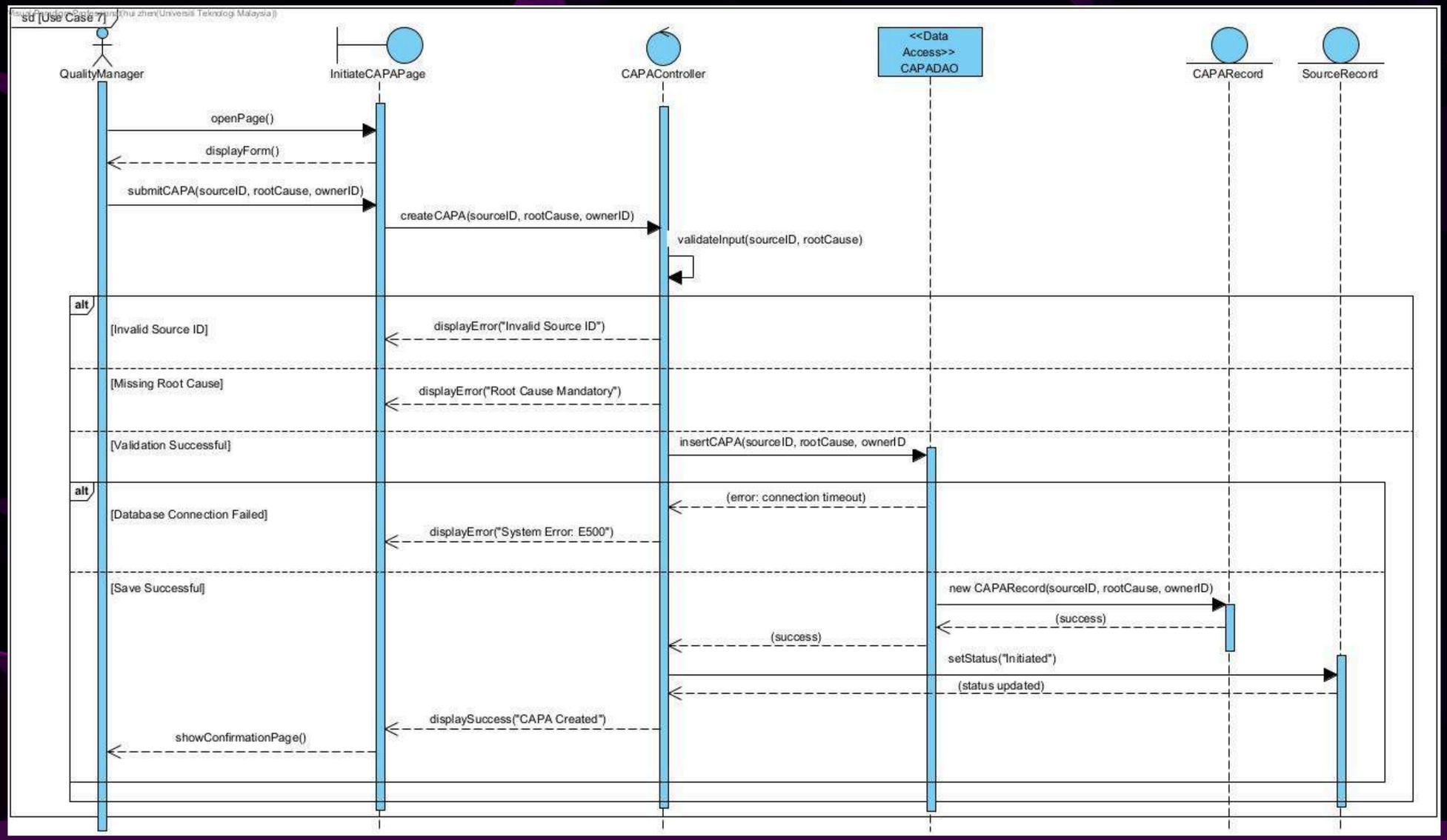
Sequence Diagram: SD003



Sequence Diagram: SD006



Sequence Diagram: SD007



THANK YOU